\

INCH-POUND
MIL-PRF-28876/9F
1 March 2021
SUPERSEDING
MIL-PRF-28876/9E
4 October 2004

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, FIBER OPTIC, CIRCULAR, PLUG STYLE, MULTIPLE REMOVABLE TERMINI, SCREW THREADS, WITH 90-DEGREE BACKSHELL, ENVIRONMENT RESISTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-28876.

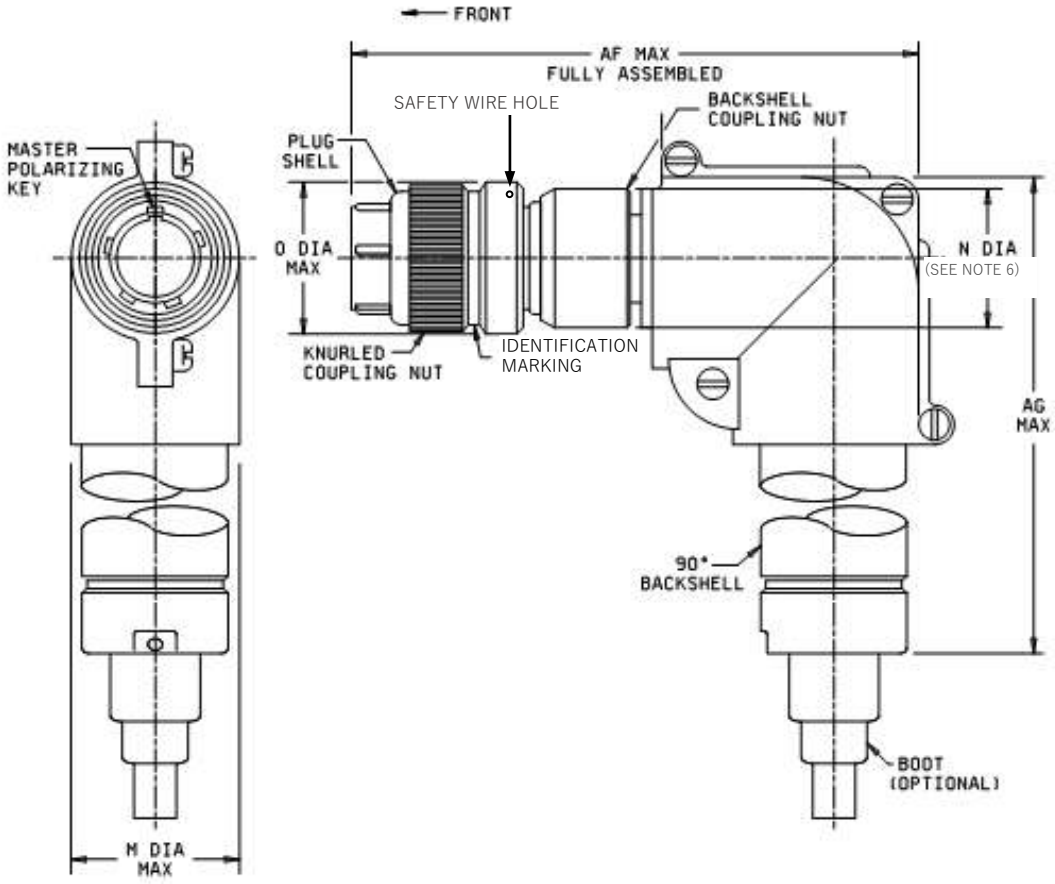


FIGURE 1. Plug connector with 90-degree backshell.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:31Z
Check the source to verify that this is the current version before use.

Shell size	M dia max	AF max	AG max	O dia max	Insert figure 1/
11	1.200 (30.48)	4.190 (106.43)	4.250 (107.95)	1.028 (26.11)	B-1
13	1.200 (30.48)	4.190 (106.43)	4.250 (107.95)	1.141 (28.98)	B-2
15	1.550 (39.37)	4.440 (112.78)	4.500 (114.30)	1.263 (32.08)	B-3 or B-4
23	2.250 (57.15)	4.850 (123.19)	5.000 (127.00)	1.705 (43.31)	B-5 or B-6

FOOTNOTE:

1/ For insert arrangement, see the indicated figure in Appendix B of MIL-PRF-28876.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents (mm) are given in parentheses for general information only.
3. Dimensions apply to plated/finished part.
4. Metric equivalents (mm) are in parentheses.
5. Identification marking shall be in accordance with 3.10.3 of MIL-PRF-28876.
6. See appendix A of MIL-PRF-28876 for N DIA dimensioning.
7. Safety wire holes shall be in accordance with MIL-PRF-28876.

FIGURE 1. Plug connector with 90-degree backshell - Continued.

REQUIREMENTS:

Dimensions and keying configurations. See figure 1 herein and figures A-1, A-3, A-5, A-6, and A-7 of MIL-PRF-28876.

Weight (without termini).

Shell size	Weight, ounces (grams) max
11	9.5 (269.325)
13	11.5 (326.025)
15	14.5 (411.075)
23	27.0 (765.45)

Protective caps. Each connector shall be supplied with protective caps over the front and rear portions of the connector.

Insert arrangement. See Appendix B of MIL-PRF-28876 for desired shell size.

Termini. Style P (pin) or S (socket) (see MIL-PRF-29504/14 and MIL-PRF-29504/15). (For dummy terminus, see MIL-PRF-29504/3.)

Shell polarization. 1 through 6 keyway positions (see figure A-3 of MIL-PRF-28876).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:31Z
Check the source to verify that this is the current version before use.

Part or Identification Number (PIN). Marked on coupling nut of plug (see table I).

M	**28876**	**/9**	**X**	**X**	**X**	**X**	**X**	**X**
Prefix for military specification	Specification number	Spec sheet no.	Shell size designator (see table I)	Insert arrangement number (see table I)	Backshell dash number (see table I)	Terminus/ insert style (see table I)	Key position number (see table I)	Class designator (see table I)

TABLE I. PIN designators.

Shell size	Shell size designator	Insert arrangement number	Backshell dash number (see table II)	Terminus/ insert style	Key position	Class designator (see table III)
11	A	1	1 or 2	P or S	1, 2, 3, 4, 5, or 6	T or blank
13	B	1	1, 2, or 3	P or S	1, 2, 3, 4, 5, or 6	T or blank
15	C	1 or 2	1, 2, or 3	P or S	1, 2, 3, 4, 5, or 6	T or blank
23	F	1 or 2	1, 2, or 3	P or S	1, 2, 3, 4, 5, or 6	T or blank

TABLE II. Backshell dash number.

Shell size	Shell size designator	Maximum cable outer diameter in accordance with backshell dash number inch (mm)		
		1	2	3
11	A	0.250 (6.35)	0.346 (8.79)	--
13	B	0.285 (7.24)	0.346 (8.79)	0.453 (11.50)
15	C	0.500 (12.70)	0.250 (6.35)	0.375 (9.53)
23	F	0.866 (22.00)	1.000 (25.40)	0.600 (15.24)

TABLE III. Class designators.

Class designator	Description
Blank	Aluminum with non-conductive finish
T	Aluminum with conductive plating

Example: M28876/9B12P1

Fiber optic cable diameter. See table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:31Z
Check the source to verify that this is the current version before use.

Mating counterpart. Receptacle connectors are specified in MIL-PRF-28876/1 through MIL-PRF-28876/5 and MIL-PRF-28876/11 through MIL-PRF-28876/14. Dust covers are specified in MIL-PRF-2886/10.

Installation and removal tools. As specified in Drawing 499-6872813.

For qualification, conformance, and periodic inspections, dummy termini shall be used in all unused cavities.

NOTES:

Referenced documents. In addition to MIL-PRF-28876, this specification sheet references the following documents:

MIL-PRF-28876/1	MIL-PRF-28876/2	MIL-PRF-28876/3
MIL-PRF-28876/4	MIL-PRF-28876/5	MIL-PRF-28876/11
MIL-PRF-28876/12	MIL-PRF-28876/13	MIL-PRF-28876/14
MIL-PRF-29504/3	MIL-PRF-29504/14	MIL-PRF-29504/15
Drawing 6872813	MIL-PRF-28876/10	

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA – CC

Preparing activity:
Navy − SH
(Project 6060-2014-032)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T20:31Z
Check the source to verify that this is the current version before use.